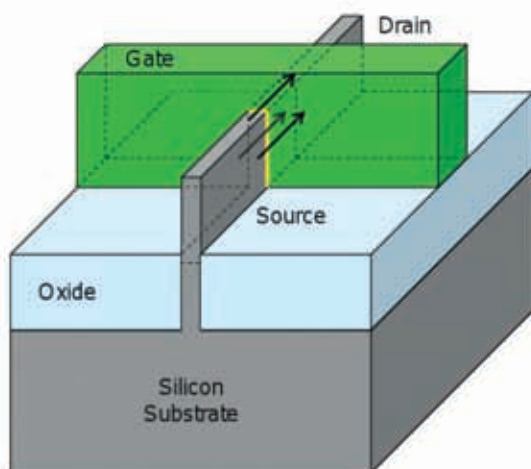


Out numbered!

Intel factories produce over 5 billion transistors every second. That's 150,000,000,000,000,000 transistors per year, the equivalent of over 20 million transistors for every man, woman and child on earth.

More than 6 million 22-nm tri-gate transistors could fit into a single full stop.

22nm Tri-Gate Transistor



3-D Tri-Gate transistors form conducting channels on the three side of a vertical fin structure, providing "fully depleted" operation. Transistors have now entered the third dimension!

it larger by expanding its Z-axis.

3D TRANSISTORS: When more is Moore, err more

Paul Otellini, Intel's President & CEO says: "Intel's scientists and engineers have once again reinvented the transistor, this time utilizing the third dimension. Amazing, world-shaping devices will be created from this capability as we advance Moore's Law into new realms."

Intel refers to their Tri-gate transistor

a "fully depleted" one. As you can see from the diagram above, there is a large increase in surface area of the gate that is in contact with the conductive surface that is also called an "inversion layer", because it can conduct and insulate. On account of the greater surface area and better current flow, there is also a clearer disparity between on and off states of the transistor, and therefore, the transistor can now be built to switch on and off even faster. This has a directly favourable

impact on clock speeds, which is something we've seen largely stagnate over the past 2 years, with only fractional increments.

This also brings about power savings. Applying less voltage to the gate will reduce the conductivity of the inversion layer, but owing to the increased surface area it'll still carry across the same number (or even more) electrons. Even better, now that Intel has used this finned design, the size of the fin can be increased in the future. For now though, Intel's new design can reap the benefits of higher clock speeds and power savings and these are not mutually exclusive.

What does this mean?

Intel claims these tri-gate transistors in their upcoming 22nm Ivy Bridge processors can switch between on and off 18 to 37 per cent faster than Sandy Bridge (32nm) processors. That number can directly equate to increased clock speeds. In terms of power saving, Intel mentions a figure in the range of 50 per cent, which is huge. This has large implications on not just desktop and notebook processors. Think ARM competition, think Intel in handhelds and smartphones, and while there's no guarantee that this advance will see them there, but it's surely a big milestone on the way and one that widens the gap between them and the competition. **d**

We had a short chat with R. Ravi-chandran, Director-Sales, Intel South Asia and put forth a few generic questions:

Q: Is this going to revolutionise CPU architectures – dramatically more cache, higher clock speeds to the tune of 4+ GHz for quad cores?

A: Intel's 3-D Tri-Gate transistors enable chips to operate at lower voltage with lower leakage, providing an unprecedented combination of improved performance and energy efficiency compared to previous state-of-the-art transistors. These capabilities give chip designers the flexibility to choose transistors targeted for low power or high performance, depending on the application. The performance can be achieved fundamentally

by CPU architecture, in combination with cache/architecture/size, and clock speed. For example moving to 22nm on Tri-gate Transistors will increase performance approximately by about 37% and active power reduction by about 50%.

Q: How is a Tri-gate transistor exciting for a consumer? Processors are a back-end device, and viewed as generally boring. I mean you buy a laptop, not a processor. In a line, why should people get excited?

A: Yes, you buy a laptop and it is primarily powered by a CPU – Central Processing Unit. Intel's 3-D Tri-Gate transistors enable CPUs to operate at lower voltage with lower leakage, providing an unprecedented combination

of improved performance and energy efficiency compared to previous state-of-the-art transistors, that will translate to end user needs like new sleeker form factors, increased battery life etc.

Q: Are we going to see dramatically cheaper options in the entry-level segment?

A: Intel traditionally introduces a new process on the core business products (PCs and servers) that have typically driven the majority of our silicon volume and capacity, allowing quick ramp and high utilisation of the fabs. As we look forward to future process generations, we will evaluate the silicon volume and capacity situation depending on the market conditions and customer needs.